



Income insufficiency impacts early brain development in infants facing increased psychosocial adversity: A network-based approach

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Early adversity is associated with brain changes that negatively impact development, social-emotional functioning, and academic achievement. The multifaceted nature of adversity complicates efforts to isolate specific factors that affect brain development and developmental outcomes. In the present study, we leverage parent survey data and longitudinal electroencephalography (EEG) data collected from infants during 4-, 9-, and 12-mo well-child visits at a large hospital-based urban primary care clinic serving predominantly low-income families (293 infants, 667 EEGs). Using a network framework, we aimed to identify specific socioeconomic and psychological factors associated with differences in infant brain development. We find that mothers who report to be income insufficient were more likely to have lower educational attainment, report low-income, experience higher levels of stress, and encounter more adverse life events. Controlling for these variables, income insufficiency was specifically associated with delayed brain development in the first year of life; infants developing in a household where parents felt their income was inadequate to support the family's needs exhibited slower rates of change in alpha peak frequency, alpha power, and beta power. Together, these findings provide a framework of understanding and visualizing how early adversity impacts neurodevelopment and provides evidence for the potential utility of maternal income sufficiency as an additional screening tool to accelerate identification of populations most vulnerable and in need for early intervention.

EEG | adversity | income sufficiency | infancy | neurodevelopment

In the first year of life, the brain undergoes rapid growth, with accelerated synapse formation and the establishment of functional neural circuits. These rapid changes are associated with heightened plasticity, allowing the brain to become uniquely primed to take in information, and in turn, form new circuits. Thus, early experience can have lasting effects on cognitive, emotional, and social functioning, highlighting both the adaptability and vulnerability of brain development early in life (1–3). Identifying factors that contribute to vulnerability early in development is critical for improving developmental outcomes.

It has now been extensively documented that early psychosocial adversity (e.g., exposure to elevated levels of stress, intimate partner violence, maltreatment) place infants at increased risk for mental health and developmental disorders that manifest later in life (2, 4, 5). Beyond these developmental outcomes, alterations in brain activity have been associated with family income (6–8), maternal stress (9, 10), neighborhood disadvantage (11), or severe forms of neglect (12). Differences in the patterns of brain activity associated with early psychosocial adversity have been found to persist into childhood, impacting later cognitive development (13, 14) and educational achievement (15–17). In addition, recent findings suggest that early forms of adversity (such as living in poverty) accelerates the *trajectory* (18) or *pace* (19) of early brain development and may lead to poorer cognitive outcomes in toddlerhood (19).

While there is strong evidence that poverty has detrimental effects on child development, the complexity of this multifaceted construct poses significant challenges for early screening of risks for developmental disorders and the development of targeted interventions. Measures of socioeconomic status (SES) are complex and interconnected, encompassing factors such as financial hardship, access to resources, parental education, and occupational prestige (4), as well as health-related factors such as insurance and caregiver psychopathology. Most large research studies examining SES and brain development

Significance

This paper demonstrates the key role of adversity on the earliest stages of postnatal brain development. Data from the Baby Steps study, an ongoing longitudinal study collecting electroencephalography (EEG) and socioeconomic status related data from infants attending a primary care clinic serving families with elevated rates of poverty and stress, demonstrate that infants developing in a household where mothers reported income to be inadequate to support the family's needs exhibit delays in brain maturation in the first year of life. Maternal perceptions of income sufficiency may signal heightened risk for neurodevelopmental delays along with increased financial and psychosocial stress. Our network approach highlights the potential for a screening tool to identify vulnerable infants early and guide pathways for targeted interventions.

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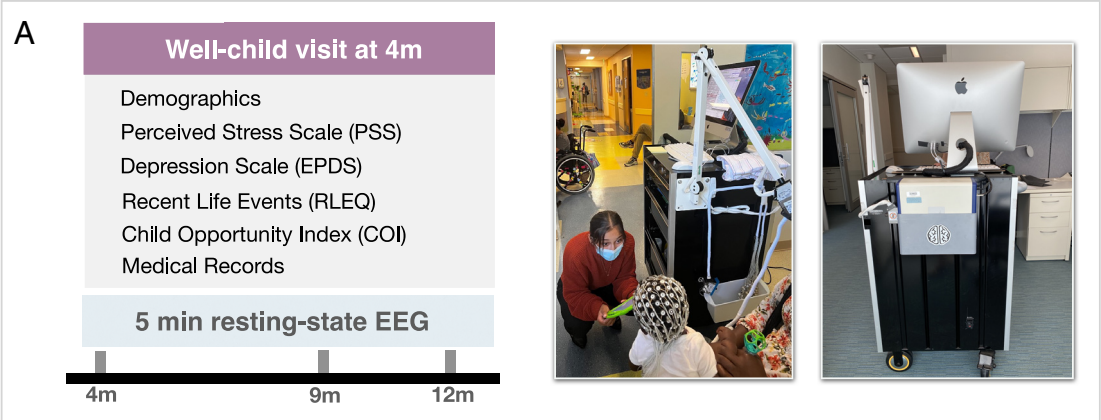
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have studied populations with a broad range of SES, resulting in high collinearity between SES-associated factors, making it difficult to isolate the impact of specific factors on brain development and developmental outcomes. To date, limited empirical work has employed the longitudinal data necessary to model associations between psychosocial factors and early brain trajectories, particularly in infant populations experiencing the extremes of adversity.

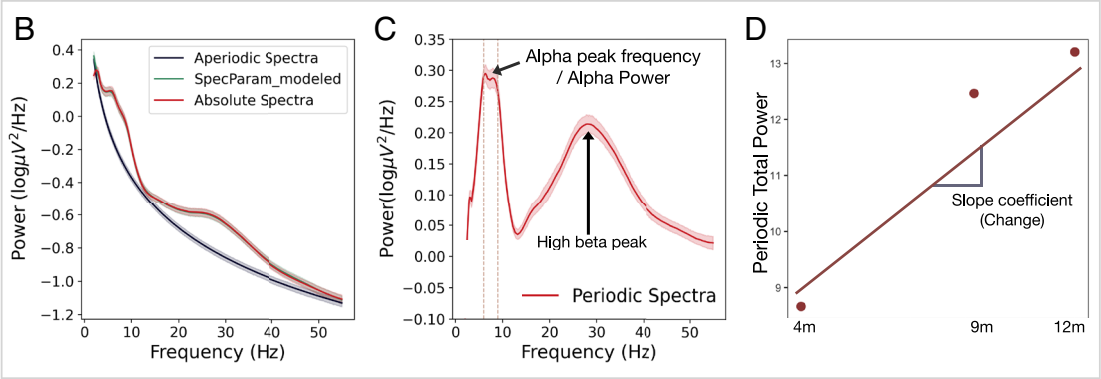
The focus of this paper was to identify psychosocial factors most strongly associated with altered early trajectories of brain activity during the first year after birth within a community-based sample of infants living in high rates of adversity. We utilize data from an

on-going study (*Baby Steps*), which is a prospective, longitudinal electroencephalography (EEG) study embedded in an urban hospital-based primary care clinic that primarily serves low-income families. The ultimate study aims are to identify factors during the first year of life that are predictive of autism or developmental delay assessed at 2 y of age. At 4-, 9-, and 12-mo well-child visits, resting state EEG is collected in the clinic along with surveys completed by parents that include measures of SES, parental stress, and resource-related questions (Fig. 1A). To address the current aims of this paper, we used Exploratory Graph Analysis (EGA; Fig. 1E) to analyze the structure of a multitude of factors associated with SES and stress and aimed to identify the factors that are

Data Collection



EEG Computation



Network Analysis

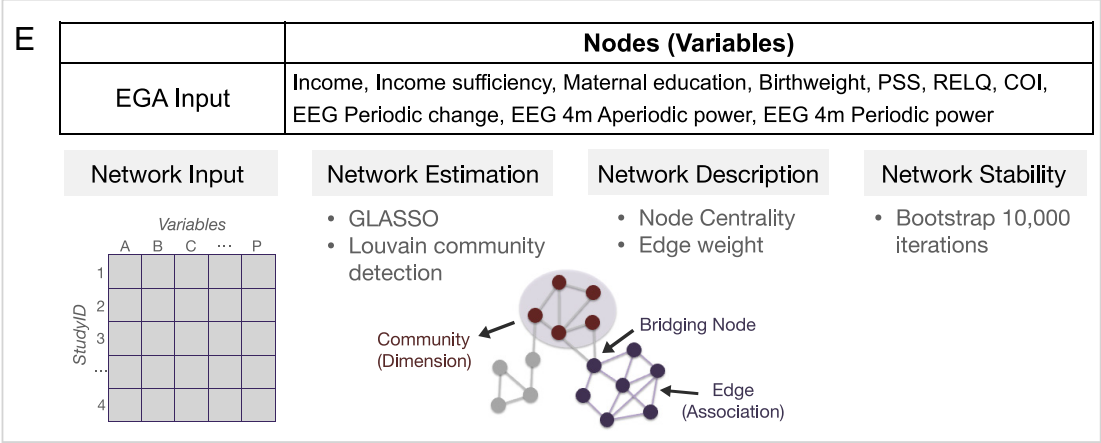


Fig. 1. (A) Left: Diagram of data collection at well-child visits (4, 9, and 12 mo) at the primary care clinic. Right: Photo during an EEG data collection and the portable EEG Cart. (B) Example of absolute power spectra and aperiodic power spectra represented as the mean value with shaded errors describing 95% CI. Aperiodic total power is derived by computing integral under the aperiodic spectra from 2 to 50 Hz (navy). (C) Example of aperiodic-adjusted periodic spectra, dashed lines representing alpha frequency band (6 to 9 Hz). Periodic total power is derived by computing integral under the periodic spectra (pink). (D) Example of an estimated linear slope coefficient of periodic total power from an individual to derive change in EEG measures. (E) Summary of items included in the final EGA, and a schematic representation of the workflow of network structure estimation (to construct the network), network description (to characterize the network), and network stability analysis (to assess the robustness of results) following Borsboom et al. (20).

most strongly associated with early trajectories of brain activity (two global estimates of brain activity during the first year of life in a set of 293 infants (See [SI Appendix, section SI1](#) for details on Baby Steps Study design. [SI Appendix, Fig. S1](#) includes a diagram of the number of completed visits, EEG data acquisition, and usable data in the current study).

Results

Background questionnaires, almost all of which were completed by mothers (see [Table 1](#) for descriptive information on demographic and survey responses) confirmed high levels of economic hardship in our sample with 71.7% reporting being publicly insured, close to 60% living below the federal 200% poverty line (58% families reporting monthly household income less than \$4,400), 68.6% earning below bachelor's degree, and more than 50% living in single-parent households. Furthermore, survey responses indicated moderate to high levels of stress for many families in the study, with 24.1% reporting “never” or “rarely” having sufficient income to meet their family’s needs, around 40% reporting moderate to high levels of stress on the Perceived Stress Scale (PSS), and 43% at or below “Low” levels of child opportunity index (COI), standardized on a national level.

Here we used EGA to assess how parent-completed questionnaires assessing stress and SES-associated factors cluster together (i.e., complementary to latent factors in a traditional exploratory factor analysis) and identify which variables play pivotal roles in impacting *changes* in early brain development. EGA is a statistical method commonly used in psychometric and behavioral studies that can identify and visualize the multidimensionality of datasets ([20, 21](#)). Data are represented in a network including central nodes (variables), communities (i.e., dimensions; groups of closely related variables), and edges (pathways connecting variables). A key strength of EGA lies in its ability to detect the existence and weight of edges, which reflect the *conditional associations* between variables—statistical relationships between two variables that remain after taking into account all other variables in the data (e.g., partial correlations). EGA removes weak and multicollinear associations using penalized maximum likelihood estimation ([22](#)), resulting in the disconnection of such nodes in the network representation. Consequently, strong and unique associations remain in the network structure. The EGA included 293 participants who met the following criteria: 1) provided EEG data of good quality at one or more time points ([SI Appendix, Fig. S1](#)) and 2) completed the demographic and survey questionnaires. See [SI Appendix, Table S4](#) and [Fig. 1E](#) for a summary of the EGA model final input and network analysis workflow. Given the limited sample size and multicollinearity across frequency band EEG measures, we limited the EGAs to two global estimates of brain activity—aperiodic and periodic total power (across 2 to 55 Hz; [Fig. 1 B and C](#)) and the slope of change for these measures within each participant was included ([Fig. 1D](#)). We focused on frontal EEG power based on past findings relating early life adversity and brain development ([13, 18](#)) and frontal EEG and later functioning ([23](#)).

Income Sufficiency Serves as a Central Node That Bridges the 3 Dimensions (SES, Stress, and EEG). EGA resulted in a low-density sparse network (edge density of 0.27) that occurs when there are selective, but strong relationships within the network ([Fig. 2A](#)). The network structure is comprised of 3 expected dimensions: dimension 1 (blue) consists of SES-related variables (household income, maternal education, COI); dimension 2

Table 1. Demographic (N = 293) and survey responses

Sex	155 M, 138 F
Birth weight [Mean (range)]	3.35 (2.27-4.82)
Race, n (%)	
African American or Black	114 (38.9)
Asian	2 (0.7)
Native American	7 (2.4)
Native Hawaiian or Pacific Islander	1 (0.3)
White	26 (8.9)
More than one	42 (14.3)
Not answered	101 (34.5)
Ethnicity, n (%)	
Hispanic or Latino/a	177 (60.4)
Not Hispanic or Latino/a	115 (39.2)
Insurance type, n (%)	
Medicaid	210 (72.0)
Private	82 (28.0)
Child Primary Language, n (%)	
English	168 (57.3)
Spanish	50 (17.1)
English and Spanish	75 (25.6)
Monthly household income, n (%)	
<\$2,100	81 (27.6)
\$2,101-\$4,400	89 (30.4)
\$4,401-\$7,400	43 (14.7)
\$7,401-\$12,700	34 (11.6)
>\$12,700	45 (15.4)
Not answered	1 (0.3)
Marital Status, n (%)	
Single	140 (47.8)
Cohabiting	50 (17.1)
Married	91 (31.1)
Separated, divorced, widowed	12 (4.1)
Maternal Education, n (%)	
<8th grade	8 (2.7)
Some High school	21 (7.2)
High school/GED	78 (26.6)
Associate's degree/some college	94 (32.1)
>Bachelor's degree	92 (31.4)
Income sufficiency, n (%)	
Never	40 (13.7%)
Rarely	31 (10.6%)
Sometimes	99 (33.8%)
Usually	69 (23.5%)
Always	54 (18.4%)
Employment status, n (%)	
Never employed	5 (1.7)
Currently Unemployed	90 (30.7)
Part-time	42 (14.3)
Full-time	154 (52.6)
Perceived stress scale, n = 241	
mean (SD)	11.76 (7.76)
Low, n (%)	139 (60.2%)
Moderate-High (above 14), n (%)	92 (39.8%)
Depression, n = 223	
Mean (SD)	5.07 (5.45)
Above clinical cutoff, n (%)	24 (10.1%)

Table 1. (Continued)

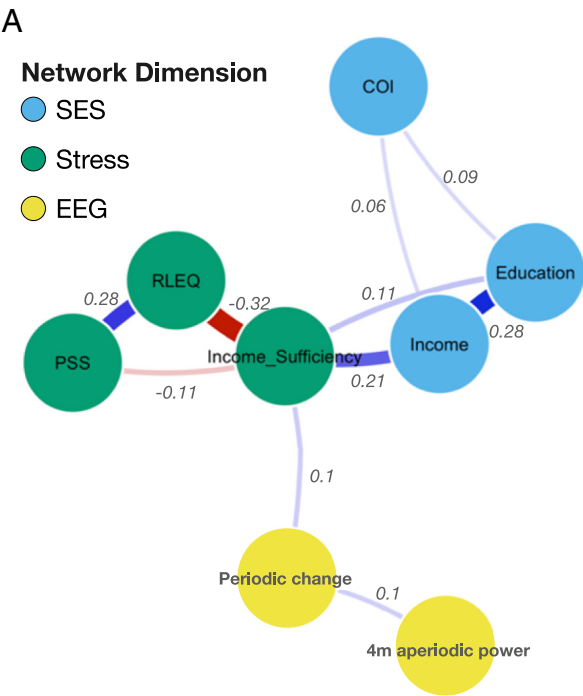
Recent life events, <i>n</i> = 230		
Events occurred 0 to 4mo, Mean (SD)	1.27	(1.8)
Child opportunity index, <i>n</i> = 281		
Overall COI (z-score) mean (SD)	0.21	(0.68)
Very Low, <i>n</i> (%)	34	(12%)
Low, <i>n</i> (%)	87	(31%)
Moderate, <i>n</i> (%)	36	(12.8%)
High, <i>n</i> (%)	73	(25.9%)
Very High, <i>n</i> (%)	51	(18.3%)

(green) represents stress-related variables [perceived stress (PSS), recent life events (RLEQ), and income sufficiency]; and dimension 3 (yellow) consists of EEG measures (periodic slope estimate and 4 m aperiodic power). Birthweight and 4 m periodic power did not belong to a dimension.

The stability of the network structure was validated using Bootstrap EGA (24), which resulted in a median of three dimensions (Dim; SD: 0.57; CI: 1.9, 4.1), and moderate-to-high levels of structural consistency (Dim 1: 0.76, Dim 2: 0.74, and Dim 3: 0.64). The bootstrapped dimension frequency is reported in *SI Appendix, Table S5*, and *Fig. 2B* displays the bootstrapped item stability loading across each dimension. Income sufficiency, RLEQ, and income nodes were most central (i.e., high centrality) in the network based on having the most connections with other nodes. Further assessment of the centrality stability found that income sufficiency exhibited the highest strength centrality, indicating it serves an influential role in the network structure (*SI Appendix, Figs. S5 and S6*)—including strong associations with both variables in SES-related and stress-related dimensions. The robustness of edge associations was further validated using

bootstrap analysis and replication with unregularized Graphical Markov Models with Mixed Graphs (*SI Appendix, Figs. S7 and S8 A and B*).

Income Sufficiency Reflects Stress Related to Economic Hardships. To obtain a more nuanced understanding of income sufficiency, we examined the nodes identified to be associated with income sufficiency (simplified to three ordinal categories: 1- Never, combining Never and Rarely, 2- Sometimes, and 3- Always, combining Always and Usually) in dimensions 1 (Income, Education) and 2 (PSS, RLEQ). First, significant differences in the distribution were observed in income [$\chi^2(2) = 54.45, P < 0.001$] and education [$\chi^2(2) = 21.74, P < 0.001$] grouped by income sufficiency (*Fig. 3A*; see *SI Appendix, Table S6* for a full table). Post hoc comparisons with Benjamini–Hochberg correction indicated significantly lower income in mothers who reported to be “Never” vs. “Always” ($P\text{-adj} < 0.001$), and significantly lower education attainment in the “Never” compared to the “Sometimes” ($P\text{-adj} = 0.01$) and “Always” ($P\text{-adj} < 0.001$) income sufficient groups (see *SI Appendix, Table S7* for post hoc pair-wise comparisons of income sufficiency and SES-related demographic variables). To explore the associations in dimension 2, Item Response Theory (IRT) analysis was conducted. IRT produces a distribution of theta (θ) values, where higher theta scores indicate increased levels of the underlying latent trait assessed via surveys. IRT findings (with Benjamini–Hochberg correction) in RLEQ and PSS suggested that mothers who reported the income to be Never sufficient exhibited significantly higher theta, indicating significantly elevated levels of stress and exposure to recent life events as compared to mothers who reported to be Sometimes or Always income sufficient (*Fig. 3B* and *SI Appendix, Figs. S11 and S12* reports item-level endorsements of PSS and RLEQ). Furthermore, we conducted item-level responses on both the RLEQ and PSS using Differential



Items	Dim 1	Dim 2	Dim 3
Income	0.994		
Income Sufficiency	0.229	0.748	
Education	0.994		
PSS		0.986	
RLEQ		0.988	
Periodic change		0.248	0.636
4m aperiodic power	0.103	0.136	0.675
COI	0.742		0.153

Fig. 2. (A) Network structure from EGA (*n* = 293). Circles indicate nodes (variables), and lines indicate edges (association) between nodes (blue: positive association, red: negative association). Thicker lines indicate greater magnitude of the association (Edge-weights are reported alongside the edges). EGA indicates a 3-dimension solution: SES-related variables (household income, maternal education, COI); Stress-related variables (PSS, RLEQ, and income sufficiency); and EEG measures (periodic slope estimate and 4 m aperiodic power). (B) Item stability across all bootstrap replicate samples. Standardized loadings $\geq |0.10|$ are reported. Abbreviations: PSS—Perceived Stress Scale, RLEQ—Recent Life Events Questionnaire, COI—Community Opportunity Index, Dim—Dimension.

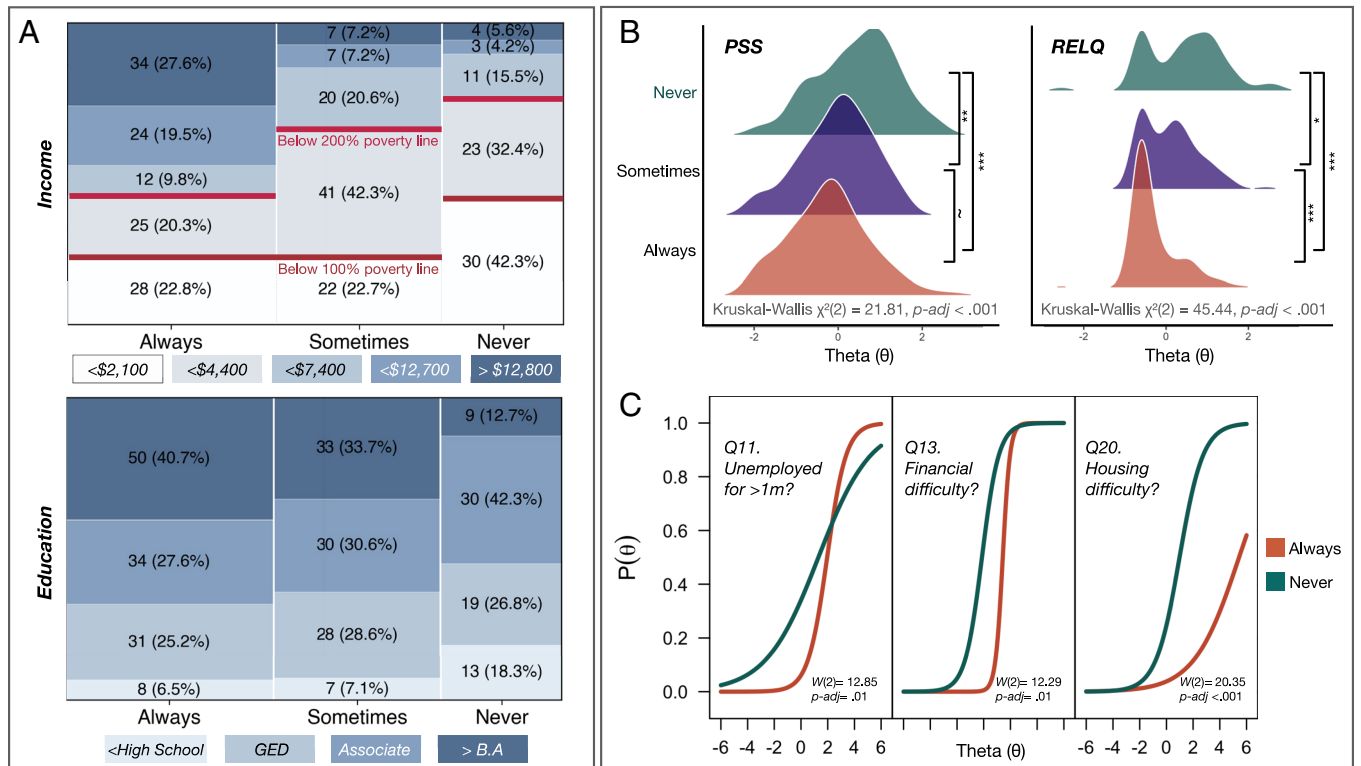


Fig. 3. (A) Mosaic plot of reported household income and maternal education levels grouped by income sufficiency ($n = 292$). (B) Density plot of the estimated IRT theta distribution (Left: PSS, Right: recent life events) marked with significance indicating differences between groups (Dunn's test with Benjamini-Hochberg correction was conducted following a significant Kruskal-Wallis test), and (C) Significant item-level DIF in the RLEQ (items 11, 13, and 20) between Never and Always income sufficient groups. Theta on the x-axis indicates latent traits on the survey, and the y-axis indicates the probability of responding to the item with a 1 (i.e., with a "yes"). The item characteristic curves derived from DIF indicate that the Never income sufficient group is more likely to endorse "yes" to these items. For example, given latent trait = 0 (on the x-axis), the probability of responding with "yes," or in this case "1," for an individual from the Never income sufficient group is significantly higher than the Always group.

Item Functioning (DIF) analysis with Benjamini-Hochberg correction to verify which items were disproportionately associated with reporting to be Never income sufficient. On the RLEQ, three items, all related to financial hardships (i.e., unemployment for more than 1-mo, financial difficulty in paying bills, and housing instability) showed significant DIF between the Never and Always income sufficient groups (Fig. 3C), suggesting these items disproportionately contributed to reporting lower income sufficiency. No significant DIF were identified based on item-level responses between Never and Sometimes income sufficient groups for the RLEQ nor for any items on the PSS based on income sufficiency. Together, our network findings suggest that mothers reporting to have insufficient income tend to exhibit higher levels of stress, likely driven by economic and material disparity.

Income Sufficiency Is Positively Associated with the Rate of Change in Brain Activity in the First Year of Life. Income sufficiency, but no other environmental variables, was found to be positively associated with change in brain activity (i.e., change in periodic total power). The positive association between income sufficiency and developmental changes in brain activity suggests that the infants of mothers who reported to have inadequate income were more likely to show slower rates of change in periodic total power over the first year of life. Periodic spectra at 4, 9, and 12 mo grouped by income sufficiency showed reductions in both the alpha (6 to 9 Hz) and high beta (20 to 30 Hz) periodic power as early as 9 mo (Fig. 4A; see SI Appendix, Figs. S13 and S14 for EGA findings and EEG power spectra based on central and posterior regions). Linear mixed-effects models revealed significant

interactions between age and income sufficiency for both periodic alpha power (standardized $\beta = 0.08$, fdr-adjusted $P = 0.02$) and alpha peak frequency (standardized $\beta = 0.08$, fdr-adjusted $P = 0.01$) (See SI Appendix, Table S8 for full output). Specifically, the "Never" income sufficient group showed a markedly altered developmental trajectory, with slower rates of increases in alpha power (Fig. 4B) and alpha peak frequency (Fig. 4C). In addition, a main effect of income sufficiency was found for beta power, (standardized $\beta = 0.14$, fdr-adjusted $P = 0.01$), with the "Never" sufficient group demonstrating consistently lower beta band power throughout the first year relative to the other groups (Fig. 4D).

Discussion

In our longitudinal sample drawn from a primary care clinic serving families disproportionately experiencing poverty, mothers who report to be income insufficient were more likely to have lower educational attainment, report low-income, report higher levels of stress, and experience more adverse life events. Controlling for these variables, we identified specific associations between income insufficiency and infant brain activity. Infants developing in a household where parents felt their income was never or rarely adequate to support the family's needs exhibited altered developmental trajectories in their brain activity (i.e., lower total periodic power) in the first year of life. Delays were particularly evident in EEG features previously shown to track with brain maturation [i.e., alpha peak frequency and beta peak activity; (25, 26)]. Together, our EGA network framework 1) provides an intuitive model that integrates insights on the dimensional structure of the

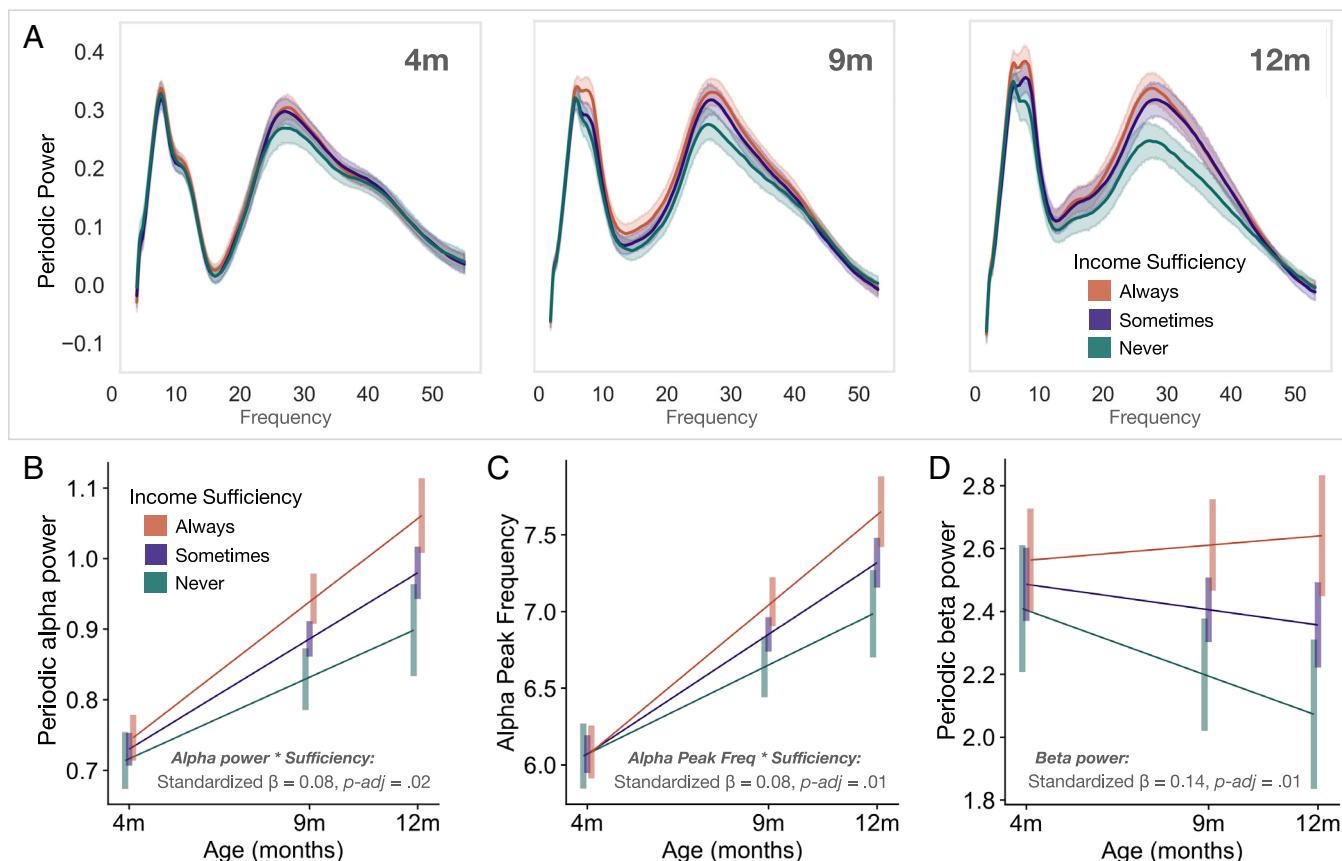


Fig. 4. (A) Periodic power spectra grouped by income sufficiency at each age visit (n : 4 m- 263; 9 m- 239; 12 m- 191). Estimated EEG measures of (B) periodic alpha power, (C) alpha peak frequency, and (D) periodic beta power grouped by income sufficiency. (Note. Error bars represent ± 1 SE. X-axis represents visit ages for visualization purposes, the models were run with age as a continuous measure).

closely interwoven SES-related and stress-related factors, instead of exploring the roles of these factors independently, 2) offers an additional screening tool for identifying at-risk populations experiencing both economic and psychological adversity, and 3) extends previous research on early impacts of psychosocial adversity, suggesting that early life income insufficiency is associated with alterations in early functional brain development.

Building on foundational social network theory and advances in network science (27), the network approach envisages clear patterns in how interconnected factors shape early development. The network analysis identified three distinct dimensions: 1) SES factors (income, maternal education, and the COI); 2) Stress factors (maternal stress, stressful life events, and income sufficiency); and 3) EEG features. Notably, income sufficiency bridged stress and SES-related variables and brain measures, underscoring its value as an indicator that integrates financial hardship and psychosocial factors impacting brain development. A mother's *perception* of how much the household income meets their needs was linked to the broad range of indicators related to economic adversity. For instance, among the mothers who reported their household income as *never* adequate, 87.3% were publicly insured (e.g., Medicaid), 74.7% fell below 200% the federal poverty line, 42.3% reported monthly incomes below \$2,100, 73.3% live in single- or separated-parent households, and close to 50% were unemployed. These mothers were also more likely to report recent financial hardships and elevated levels of stress. Our findings empirically support the well-documented finding that single parents at the lower end of the socioeconomic spectrum face accumulated risks and challenges due to limited financial resources, unstable employment, unpredictable work schedules, as well as

unstable childcare (28–30). We further extend previous findings in demonstrating that these factors have intergenerational impact, influencing the child's brain development. Of note, 43% of those who report being "Always" sufficient fall below the 200% federal poverty line, showing that sufficiency is not merely an index of income, but instead conveys distinct information beyond absolute income levels. Together, our findings provide support for the potential utility of a single question as a pragmatic screening tool that can easily be incorporated into clinical settings. While it should not replace comprehensive assessments, it may offer a practical window into early identification of the subgroup of families most likely to be experiencing multiple dimensions of stress with potential downstream risks for later developmental challenges, and thus, in need of timely interventions that could be missed or underestimated when considering factors independently.

Our finding that psychosocial adversity is associated with delayed brain development is consistent with a large body of work indicating that adversity becomes neurobiologically embedded—both structurally [e.g., differences in cortical volume, cortical thickness, and white matter integrity (31, 32, see ref. 33 for review)] and functionally [e.g., reduced EEG power; (6, 9, 34)]. Functional brain activity captured by EEG may mirror structural changes associated with early adversity. For instance, prior literature on structural brain development provides evidence that the direction of the association between SES and cortical thickness varies with age. In infancy, lower SES has been linked to thicker frontal cortex (35), potentially indicating accelerated maturation processes that narrow the window for optimal synaptic pruning, corticocortical wiring, and apoptosis (36). These neurodevelopmental changes may contribute to less synchronized neuronal

firing, reflected on the cortex as reduced oscillatory (periodic) EEG power. However, our findings of slower rate of change in brain activity is contrary to recent longitudinal infant neurodevelopmental findings (18, 19), which have suggested psychosocial adversity to be associated with accelerated brain development. Differences in the direction of findings may be attributable to differences in study populations, as our cohort primarily includes families with higher levels of adversity and lower income levels. For instance, in Wilkinson et al. (18), 15% of families reported household income of less than \$35,000 (compared to 51% in the current study), and 69% of mothers had more than a 4-y college degree (compared to 31% in the current study). Similarly, in Tooley et al. (2024), 52% of families had private insurance, vs. 28% in our study. Infants living in poverty are at elevated risk for neurodevelopmental disorders or delays, which may involve distinct underlying neurobiological mechanisms reflected in cortical brain activity. Given that our analysis is limited to changes that occur in the first year of life, longitudinal follow up will be critical to determining whether early patterns of delays persist, catch-up, or transition into accelerated trajectories during toddlerhood and early childhood, and also whether these early patterns predict longer-term behavioral and developmental outcomes.

What might slower maturational changes in EEG periodic power imply? A growing body of research has provided important links between both low- and high-frequency EEG power and later cognitive abilities. Alpha rhythms are thought to support cortical inhibition and interregional communication, playing a central role in cognition (37, 38). Higher peak alpha frequency has been suggested to be associated with stronger cognitive performance in adults (39, 40), increases with age (26, 41) in infancy, and correlates with sensorimotor development in early life (42). Alpha power measured in infancy is associated with executive function in toddlerhood (43) and high beta power (or low-gamma; 24 to 35 Hz) has been linked to early language development (7, 44). Thus, atypical or delayed maturation of EEG power may indicate delays in the development of neural circuits critical for cognitive functioning that manifest behaviorally later in childhood. As our data collection is still ongoing, particularly for the 24-mo visit when clinical outcomes can be formally assessed, we currently lack the statistical power to robustly examine associations between early EEG changes and later behavioral outcomes. Assessing these associations in a larger sample upon study completion will be essential for determining the specificity of the link between early altered trajectory and cognitive outcomes.

Of note, the EEG dimensions in the network analysis suggest positive associations between maturational changes in EEG *periodic* power and 4-mo *aperiodic* power. More specifically, infants with higher aperiodic activity at 4 mo exhibited greater changes in periodic activity over the first year of life (SI Appendix, Fig. S16). The aperiodic component of the EEG power spectrum is hypothesized to index nonoscillatory, broadband neuronal spiking activity. During early brain development, increases in the aperiodic component likely reflect growth in neuronal number, connectivity, and synaptogenesis (45). Empirical evidence suggests that aperiodic offset increases markedly during early infancy and closely tracks age throughout the first years of life (26, 41). These findings suggest that early individual differences in aperiodic activity, potentially indexing neural maturation, may shape subsequent changes in EEG periodic (oscillatory) features, important for neural synchronization. Interestingly, we did not observe a direct association between psychosocial factors and aperiodic, periodic power at 4 mo, nor birthweight. It may be important to explore features within the aperiodic component (e.g., aperiodic offset, aperiodic exponent) separately, but this question warrants further investigation in a larger sample.

Our findings of slower brain maturation in children living in households with psychosocial hardships highlight the importance of addressing the compounding challenges these families face. While our study did not focus on intervention strategies, our network findings shed light into both direct and indirect pathways through which economic adversity influences brain development. Given the high centrality of income sufficiency within the network, future work should explore the embedded associations that income sufficiency encompasses. Recent initiatives, such as Baby's First Years study (46), include directly modulating household income and exploring whether providing monthly unconditional cash transfers to mothers experiencing poverty in the first year of their children's lives gives rise to changes in infant brain activity and improves developmental outcomes. While the study is on-going, preliminary findings suggest changes in EEG high frequency power (47), though effect sizes are small and the impact on broader developmental outcomes has yet to be determined (48). Identifying the magnitude and duration of financial support needed to produce meaningful and sustained developmental change remains a significant challenge. Our network structure analysis suggests that determining a family's level of income sufficiency and what factors would successfully shift levels of income sufficiency into the "sometimes-to-always" range may be important for the success of such interventions. Alternative existing strategies that combine screening for specific unmet material needs (i.e., food, housing, and utilities), providing "in-kind" benefits [e.g., connections to Supplemental Nutrition Assistance Program (SNAP; 49)] along with approaches to mitigate psychosocial stressors related to financial hardships [e.g., job security support, access to stable child-care subsidies like Child Care Development Fund (CCDF; 50)] may also be critical components of interventions moving forward. Additionally, parenting interventions that promote responsive caregiving and coping strategies (51) may serve as an indirect, important buffer against negative developmental outcomes in children, which has been found to be beneficial in other low-middle income countries (52).

Limitation

First, while our current network approach provides a proof of concept in demonstrating the effectiveness of exploring the complexity of connectedness across different psychosocial and brain measures, it does not include other important proximal (e.g., cognitive/linguistic stimulation, parenting practices), environmental (e.g., exposure to toxins, nutrition information), nor biological factors (e.g., gene, microbiome) known to impact later child health trajectories. Future analyses should incorporate a wider range of such measures in studies with larger samples that allow assessment of "multilayer" patterns from within an individual to those at the social systems level. Second, as the study is on-going, we were unable to explore the association between the rate of brain growth to behavioral outcomes in toddlerhood. Moreover, our approach is limited in identifying protective factors within our study population. Future analyses examining buffering factors could generate valuable insights into potential interventions.

Conclusion

Taken together, the present study highlights that psychosocial factors are associated with alterations in the earliest stages of post-natal brain development. Our network approach provides a promising framework for understanding how early adversity affects neurodevelopment and for guiding prevention efforts to mitigate long-term adverse outcomes in early childhood.

Materials and Methods

Participants. The Baby Steps Study is a single-site longitudinal, prospective cohort study taking place at the Boston Children's Hospital Primary Care Clinic. The study follows infants from 4-mo to 2-y of age. Infants seen at CHPCC are recruited and enrolled prior to or during their 4-mo well child visit. Longitudinal data including 5-min resting state EEG, parent surveys, and review of the medical record are completed in the clinic at 4-, 9-, and 12-mo well-child visits. Data collection for the current study took place between February 2022–February 2025 with a total of 413 participants enrolled in the project. Exclusion criteria included gestational age younger than 36 wk; identified neurological, genetic, or clinically relevant ophthalmologic disorders; intrauterine opioid exposure, and major NICU complications, significant perinatal or neonatal complications; or family history of autism ($n = 44$). All participants had exposure to English or Spanish languages for more than 75% of waking hours and mothers were at least 18 y old at the time of the child's birth. The study was reviewed and approved by the Boston Children's Hospital Institutional Review Board (IRB-P00038922). Written informed consent was obtained from each participant's legal guardian before participation. A total of 369 were eligible to be included in the current study.

Methods. Out of 369, 331 demographics were completed (with 83% completed during the first visit—4 mo well child visit), which were filled in 98.6% by mothers (4 individuals did not disclose the relationship to the participant). See *SI Appendix, Fig. S1B* for a summary of the total number of completed demographics, surveys by study inclusion criteria.

Background and demographic information. Background and demographic information were collected from families, which included information on child and maternal race/ethnicity as well as resource-related questions (i.e., maternal education, household income). Income sufficiency was assessed with a single question: "Do you think that your household income is sufficient to meet your current family needs?" Responses were recorded as Always, Usually, Sometimes, Rarely, or Never.

Maternal reports of life events, stress, and depression.

Recent life events questionnaire (RLEQ; originally developed by ref. 53). The RLEQ is a report of stressful life events that may influence mental health. We used a modified 14-item version of the RLEQ was used to evaluate life events experienced during pregnancy or since the birth of the child. Participants were asked to indicate whether each listed event had occurred, with binary responses recorded as "Yes" or "No." Events included a range of domains such as financial difficulties, relationship changes, health concerns, and other significant stressors. The frequency of events reported to have occurred between the infant's birth and 4 mo was utilized in EGA.

PSS (54). Maternal appraisals of stress during the prior month were assessed using PSS, a 10-item scale assessing thoughts and feelings over the past month with responses rated on a 5-point Likert scale ranging from 0 (Never) to 4 (Very Often).

Edinburgh postnatal depression scale (EPDS; (55)). We used EPDS to measure postpartum depression symptoms in mothers. The scale consists of 10 items measuring thoughts and feelings experienced over the previous 7 d, with responses rated on a 4-point Likert scale ranging from 0 (Never) to 3 (Most of the Time).

COI (56). Neighborhood opportunity levels were assessed using the COI 3.0, a composite measure of 44 indicators capturing three key domains: education (e.g., access to and quality of early childhood education, high-quality schools), health and environment (e.g., availability of healthy food and green spaces), and social and economic factors (e.g., reversed poverty rates and homeownership rates) and 14 subdomains. We utilized standardized national-level COI z-scores, where a score of 0 represents average neighborhood opportunity, a COI of 1 indicates neighborhood opportunity that was 1 SD above average, and a COI of -1 indicates neighborhood opportunity 1 SD below average.

Baseline EEG.

EEG data acquisition. Two-to-five minutes of baseline, non-task-related, EEG data were acquired while infants, seated on their caregiver's lap, in a waiting area. EEG was recorded using the 128 channel HydroCel Geodesic Sensor Net (HGSN) connected to a NetAmps 400 amplifier and sampled at 1,000 Hz while infants watched a screensaver with abstract shapes for 5 min. EEG was collected with a 0.1 Hz high-pass analog filter and online rereferencing to the vertex (Cz) through NetStation software [Electrical Geodesics, Inc (EGI), Eugene, OR], and impedances were kept below 50 K Ω . Data were analyzed off-line via the Batch Automated

Processing Platform (57) with integrated Harvard Automated Preprocessing Pipeline for EEG (58). (See *SI Appendix, section S12* for more information on EEG data processing and analysis).

EEG spectral decomposition and parameterization analysis. EEG spectral decomposition was performed by calculating the power spectral density (PSD) for each 2-s segment at each electrode using multitaper spectral analysis with three orthogonal tapers. The PSD was averaged across segments for each electrode and then across regions of interest (*SI Appendix, Fig. S2*). A modified version of SpecParam v1.0.0 (Also known as FOOOF, <https://github.com/foof-tools/foof>; in Python v3.6.8) was used to model periodic and aperiodic components of the power spectra. [See Wilkinson et al. (26); *SI Appendix, Figs. S7 and S8* for more details on modifications to the original algorithm]. To characterize the aperiodic exponent (slope of the spectra), we used χ in the $1/f_x$ model fit (*Fig. 1 B and C*). Further analyses were subsequently restricted to 2.5 to 50 Hz given elevated error between 2 to 2.5, and 50 to 55 Hz. Aperiodic and periodic power across the following canonical frequency bands was calculated taking the integral of each parameterized spectra between the following frequency ranges: theta (4 to 6 Hz), low alpha (6 to 9 Hz), high alpha (9 to 12 Hz), low beta (12 to 20 Hz), high beta (20 to 30 Hz), gamma (30 to 45 Hz), and total power (2.5 to 50 Hz). Thus, aperiodic total power corresponds to the integral under the modeled $1/f$ exponent between frequency ranges 2.5 Hz (offset) and 50 Hz in linear units. To note, given that periodic power was computed as the difference between log-transformed absolute power and log-transformed aperiodic power, which is mathematically equivalent to log (absolute spectrum/aperiodic spectrum) and reflects "the relative amount of non-aperiodic (i.e., oscillatory or periodic) activity." For simplicity, we refer to this as periodic power throughout the manuscript.

Of 864 EEGs acquired (from 366 participants who participated in the EEG session across three timepoints; *SI Appendix, Fig. S1B*), 20 EEGs were not acquired, 56 EEG data were excluded due to behavioral reasons (e.g., sleeping or crying) and 31 EEG data were excluded due to poor data quality post preprocessing. A total of 743 EEGs from 348 participants provided clean EEG data (*SI Appendix, Fig. S1 C and D*).

Statistical Analysis. We used R (v4.3.2) for all our statistical analyses. The following packages were used: EGAnet package [v2.1.0 (21)]; bootnet package [v1.6 (24)]; ggplot2 [v3.5.1 (59)]; lmer [v3.1-3 (60)]; mirt [v1.43 (61)]; easyRasch package [v0.3.3 (62)].

EEG slope coefficients. Longitudinal change in frontal EEG aperiodic and periodic total power across the first year were estimated across the participants who contributed clean EEG data at least two time-points ($N = 250$ unique participants, 625 data points). EEG change (slope coefficients) from 4-to-12 mo were analyzed by fitting a linear mixed-effects model to the EEG measure, allowing for random slopes to vary. We set the 4-mo time point as our reference time point for year, such that 4-mo time point was coded as year zero. We used the coef function in the lmer package to extract model estimates of age-related change in EEG.

EGA. The final sample included in the EGAs consisted of 293 participants out of 318 who had complete demographic information and clean EEG data (*SI Appendix, Fig. S1F*). Approximately 9.3% of data was missing; however, no imputation was performed as EGA employs pairwise deletion, effectively accounting for missing data.

Dimensionality. EGA was carried out applying the graphical least absolute shrinkage and selection operator (GLASSO) and the Louvain community detection algorithm. The glasso method was estimated using a penalized maximum likelihood solution based on the extended Bayesian information criterion. Unique variable analysis was conducted using the UVA function and variables with redundant information were removed from the following analyses. The redundant items, thus, removed items, are as follows: EPDS (wto = 0.58 with PSS), aperiodic change (wto = 0.47 with 4 m aperiodic power), and marital status (wto = 0.25 with Education; see *SI Appendix, Table S4*).

Accuracy and stability. To determine the stability of the network, we conducted the bootstrapped EGA with 10,000 iterations. Structural consistency (i.e., reliability) is operationalized as the proportion of times that each dimension estimated via EGA has the same item composition across a set of replicate bootstrap samples (63). Item stability (or item replicability) indicates how often items replicate in their empirically derived dimension and in other dimensions. In addition, we conducted nonparametric bootstrapping with 10,000 samples using the

bootnet package to assess stability of the edge-estimates and centrality indices (case-drop bootstrapping).

Power analysis. Sample size simulations for the networks were implemented with powrly in R following the approach proposed by Constantin et al. (64), indicating the total sample size should be at least 218 participants for a power of 0.8. **Linear mixed-effects regression.** Linear mixed-effects regression was performed to evaluate the trajectory of periodic components (e.g., alpha peak frequency, high beta power) between income insufficiency groups over the first year of life. Income sufficiency was regarded as ordinal categorical variables (1–Never, 2–Sometimes, and 3–Always) and age at the time of EEG acquisition was included as a continuous variable. For this analysis, we used the set participants who completed demographics and provided clean EEGs ($N = 318$; SI Appendix, Figs. S1G and S1S). Separate models were fit for each EEG measure 1) periodic alpha power, 2) alpha peak frequency, and 3) periodic high beta power). Benjamini–Hochberg *fdr* correction was used to account for three models. Models were fit with the following forms:

$$\text{Model} = \text{EEG Measure} \sim 1 + \text{age (in days)} * \text{Income Sufficiency} \\ + (1 + \text{age} | \text{studyid}) + \text{sex}.$$

IRT.

Latent trait scores. We generated standardized latent trait measure for PSS, EPDS, and RLEQ using IRT. IRT theta scores of PSS and EPDS were used as input for the EGA model to account for missing responses. IRT theta scores of RLEQ was evaluated across levels of income sufficiency.

DIF. DIF was assessed across the [“Never” vs. “Always”] and [“Never” vs. “Sometimes”] groups using IRT methods. An initial logistic regression approach (RldifTable function in the easyRasch package) was used to test the presence of DIF. To evaluate DIF within an IRT framework, a two-parameter logistic (2PL) multiple-group model was fitted with SE enabled for Wald testing for dichotomous

responses (such as the RLEQ). A subsequent Wald test with Benjamini–Hochberg *fdr* correction was used to confirm the significance of DIF in items.

SI Appendix, Tables S10 and S11 summarizes the dimensionality of categorical variables that were included in each respective analysis.

Data, Materials, and Software Availability. Anonymized [csv] data have been deposited in OSF (:<https://osf.io/wc2ud/>) (65). As our data include EEG data from human infant subjects, the data shared are restricted to participants whose legal guardians provided consent for public sharing, therefore a substantial number of participants used for analyses in the paper are not included [data shared in OSF: $N = 250$ (out of $N = 317$ analyzed in the main text)]. Please contact the corresponding author with data requests of the full published sample.

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